COMP1682 Project Proposal

Integrated Study and Research Repository with Flashcard-Based Learning

Richard Egenti

BSc H COMPUTING (EXT)

001239105

Contents

1. Introduction/Overview

University students are increasingly challenged by dual demands of managing large volumes of academic material and retaining key information for long term use. As higher education shifts to digital learning environments, the volumes of notes have grown substantially through articles, blogs, books and more being available online, this has often led to disorganisation and information overload for higher education students. Sillence et al. (2022) found that while students have found various ways to manage their university-based data there are still many who experience difficulties with information overload. Similarly, Arden et al. (2024) highlights how students adapt to digital note-taking practices in lectures but also face different challenges trying to effectively integrate them into the learning routines. These findings suggest that there is a need for more effective systems that support both resource management and effective study practices.

Also, research in educational psychology consistently demonstrates that practices based on spaced repetition are among the most successful practices for improving long term retention. Varkey et al. (2024) showed that a mobile flashcard application consisting of spaced repetition significantly enhanced dental students' long-term knowledge compared to more traditional methods. Similarly, Lu et al. (2021) reported that the use of the flashcard application Anki was effective for better learning efficiency and stronger in-depth medical knowledge.

This project attempts to help higher education students overcome two typical challenges - knowledge retention and information overload. It will accomplish this by creating a web-based program that combines a research repository and a flashcard learning system. Academic materials saved in the database may be accessed directly via the platform, and each source will be connected to automatically created flashcards that emphasise key points or subjects. When students examine these flashcards, selecting a certain point directs them to the matching section of the original source. This study aims to improve the efficiency and engagement of students in organising, accessing, and retaining academic material by integrating systematic resource management with active recall.

2. Aim

The aim of this project is to develop a web-based application that combines a research library with flashcards to assist students in organising academic resources, extracting key concepts, and practicing active recall. Therefore, improving information retention, reducing overload, and encouraging better study habits.

3. Objectives

3.1 Analysis and Requirements Gathering

3.1.1 Research into existing research repository tools and flashcard systems, evaluating their functionality and limitations through academic sources. [5.0]
3.1.2 Write a literature review on the importance of information management systems and active recall methods in higher education, with clear analysis of assumptions and limitations. [12.0]
3.1.3 Research functional and non-functional requirements for an integrated repository and flashcard system, based on insights from the literature review and user expectations. [3.0]
3.1.4 Verify requirements by designing and distributing a questionnaire or short survey to students on their study habits and tool preferences. [5.0]

3.2 Design

3.2.1 Create a rich picture diagram showing students, their study workflows, and how repository and flashcard features interact. [3.0]
3.2.2 Create use-case diagrams illustrating system functionalities such as uploading resources, generating flashcards, and reviewing content. [3.0]
3.2.3 Create a conceptual ERD to outline the entities and their relationships. [3.0]
3.2.4 Create a physical ERD diagram specifying attributes, keys, and database schema structure. [6.0]
3.2.5 Design possible user interfaces using tools such as Figma or Axure RP. [7.0]

3.3 Development and Implementation

3.3.1 Implement the database schema by creating tables and relationships in MySQL or MongoDB. [10.0]
3.3.2 Develop the main user interface using a web framework using PHP and Html. [13.0]
3.3.3 Implement the resource upload and organisation features, allowing users to add, tag, and search academic sources. [8.0]
3.3.4 Implement flashcard generation linked to stored resources, displaying key concepts or topics. [13.0]
3.3.5 Create functionality were clicking a flashcard topic redirects users to the relevant section in the source. [8.0]
3.3.6 Implement user registration and login functionality with secure authentication. [8.0]
3.3.7 Add an admin panel to manage users, resources, and system data. [8.0]

3.4 Testing

3.4.1 Usability testing to observe how easily students navigate the repository and flashcard features. [5.0]
3.4.2 Black box testing to ensure system functions work as expected for users unfamiliar with the code. [5.0]
3.4.3 White box testing to verify system logic and database interactions for flashcard generation and resource retrieval. [5.0]

3.5 Evaluation

3.5.1 Evaluate whether the system meets the initial functional and non-functional requirements. [5.0]
3.5.2 Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback. [5.0]
3.5.3 Discuss limitations, challenges, and potential improvements for future development. [5.0]

4. Project Framework or Any Methodology used

The Agile methodology is adopted for this project because it supports a user-friendly approach and an extremely flexible approach to system development, which is essential for building the learning platform that integrates a research repository and flashcards. The process enables the project to move through organised but adaptive stages of design, implementation, testing, and evaluation, guaranteeing that each system component is constantly modified in response to user feedback. During the design stage, Agile facilitates the construction of user-centered interface prototypes, such as the repository dashboard, flashcard viewer, and source-linking functions, which are jointly built to meet student demands. The implementation and development stages are divided into small, manageable sprints, allowing for incremental improvements and rapid adjustments depending on testing findings. Although this project is being created independently, Scrum concepts will be modified for solo usage by considering each sprint as a concentrated development period with well-defined goals, such as design, implementation, and testing. Progress will be tracked via self-review at the conclusion of each sprint, allowing for reflection, refinement, and job prioritisation in the following phase.

Frequent testing ensures that the interface stays intuitive and that data retrieval and flashcard creation work properly. Finally, the evaluation step compares the project outputs to the original objectives, focusing on usability, functionality, and improved knowledge retention.

According to Santos et al. (2024), Agile practices such as Scrum use adaptability, collaboration and SMART project outcomes within academic environments. Similarly, Friess (2023) found that groups using Scrum achieved higher efficiency through better communication and better performance scores compared to traditional approaches. These findings support the use of agile for this project, as it allows for continued refinement to ensure that the final product meets students’ needs for information organisation and long-term knowledge retention.

5. Legal, Social, Ethical and Professional

5.1 Legal

This project must comply with all legal laws like data protection and intellectual property rules and regulations, particularly when dealing with user-generated academic work. According to Liu and Khalil (2023), compliance with the General Data Protection Regulation is critical for systems that gather or handle academic data since it provides openness and permission. According to Wolford (2025), GDPR establishes concepts such as lawful processing, data minimisation, and user control over personal information, which is critical for ensuring trust and compliance in digital academic systems. Similarly, the UK Government (2018) specifies how the Data Protection Act 2018 must maintain personal data and is protected under UK law, supplementing the GDPR's framework for responsible data management. As a result, the repository will incorporate privacy warnings and only collect data required for academic reasons. Copyright compliance will also be maintained by allowing users to contribute properly referenced academic writings that must be authorised by an administrator, preventing legal violation of published information.

5.2 Social

From a social aspect, the system encourages students to collaborate and share their experiences by also requesting permission for specific sources to be added. As highlighted by Khalil et al. (2023), educational technologies can enhance peer interaction and motivation when developed with inclusiveness with accessibility in mind. This system supports social equality by being web-based and accessible across numerous platforms, eliminating digital barriers for students from diverse backgrounds, and including accessibility features like as screen reader compatibility, configurable text size, and clear visual layouts. These design considerations enable students with visual or reading challenges to interact with academic information more freely and efficiently, ensuring that the platform is accessible to all users.

5.3 Ethical

This project prioritises academic integrity, fairness and ethical technology usage in an educational environment. According to Huang et al. (2025), ensuring transparency, privacy and accountability in digital education platforms is critical for preventing unethical activity and increasing user confidence. In compliance with these findings, the proposed repository would guarantee that all user data is anonymised and handled appropriately, while also including plagiarism prevention measures and citation prompts to promote academic integrity. Furthermore, the system’s design will ensure algorithmic fairness by preventing bias in flashcard production or source evaluation while promoting equitable learning chances for all users.

5.4 Professional

From a professional viewpoint, this project adheres to recognised computing standards for integrity, accountability, and ethical responsibility. Insights from O'Shea (2024), emphasise the importance of incorporating ethics and professional conduct into computing curricula, reinforcing the need for projects like this to adhere to ethical best practices from the start. Additionally, Blaise et al. (2024) highlights that well-structured ethical frameworks assist information systems professionals in navigating complicated problems such as privacy, data protection, and responsible system design. By incorporating these principles into the building of the system, including secure processing of academic data, fair depiction of material, and accessible design, the project displays commitment to professional standards and assures responsibility.

6. Planning (see appendix A & B)

The planning approach starts with gathering and assessing the requirements through research and user feedback, followed by the creation of diagrams and interface prototypes to visualise the repository and flashcard integration. Development will concentrate on implementing the database and core system functions in manageable stages, followed by focused testing such as usability, black box and white box testing to assure reliability and accessibility. The assessment stage will compare the final system against the project requirements and user expectations, determining how well it promotes academic resource organisation and knowledge retention.

The Agile methodology will be used to design the project, which will be completed progressively, allowing for flexibility and refinement along the process (Santos et al., 2024). Each sprint on the Gantt chart corresponds to a critical step of the project, from the analysis, design, development, testing and finally evaluation, with time estimates determined by the objectives. The nature of Agile allows for a regular self-evaluation at the conclusion of each sprint, assuring ongoing progress and compatibility with project objectives. The system will progress via several prototypes that will be adjusted based on the feedback and ensuring that each stage contributes to the final web-based solution.

6.1 Tools and Technologies

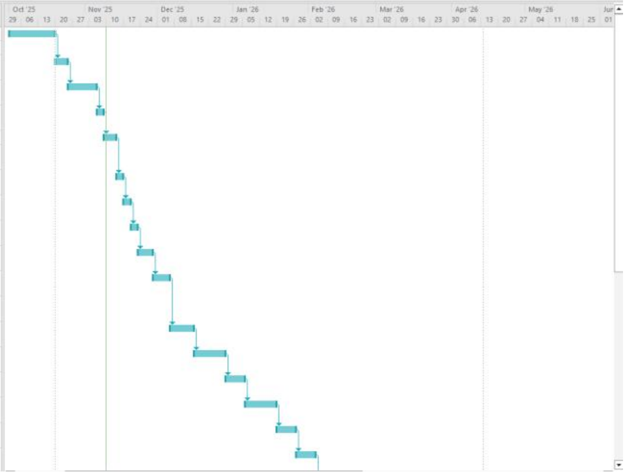
For this project I intend to use Visual Studio Code for coding with html and php to create the web-based program for my final product and all its functionalities. I will use CSS for the design of the web-based program and have some JavaScript elements for the interface and some of the features of the program. I will use either MongoDB or MySQL to create the back-end database. The design diagrams will be created using Lucidchart which will accommodate the proper creation of objects and their relationships.

7. Appendices

A.

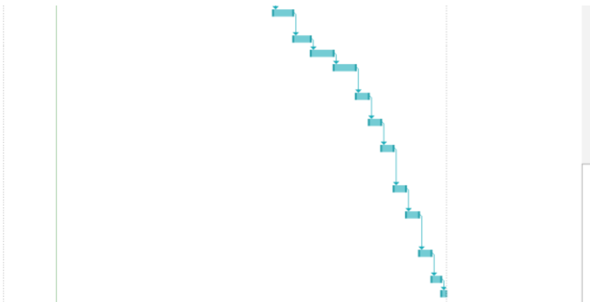
GANTT CHART

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Planning for Final year project and proposal	19 days	Wed 01/10/25	Sun 19/10/25	
2		**3.1 Analysis and Requirements Gathering**				
3		3.1.1 Research into existing research repository tools and flashcard systems	5 days	Mon 20/10/25	Fri 24/10/25	1
4		3.1.2 Write a literature review on the importance of information management systems	12 days	Sat 25/10/25	Wed 05/11/25	3
5		3.1.3 Research functional and non-functional requirements for an integrated repository and flashcard system	3 days	Thu 06/11/25	Sat 08/11/25	4
6		3.1.4 Verify requirements by designing and distributing a qu	5 days	Sun 09/11/25	Thu 13/11/25	5
7		**3.2 Design**				
8		3.2.1 Create a rich picture diagram showing students, their study workflows.	3 days	Fri 14/11/25	Sun 16/11/25	6
9		3.2.2 Create use-case diagrams illustrating system functionalities	3 days	Mon 17/11/25	Wed 19/11/25	8
10		3.2.3 Create a conceptual ERD to outline the entities and their relationships.	3 days	Thu 20/11/25	Sat 22/11/25	9
11		3.2.4 Create a physical ERD diagram specifying attributes, keys, and database schema structure.	6 days	Sun 23/11/25	Fri 28/11/25	10
12		3.2.5 Design possible user interfaces using tools such as Figma or Axure RP.	7 days	Sat 29/11/25	Fri 05/12/25	11
13		**3.3 Development and Implementation**				
14		3.3.1 Implement the database schema by creating tables and relationships in MySQL or MongoDB.	10 days	Sat 06/12/25	Mon 15/12/25	12
15		3.3.2 Develop the main user interface using a web framework using PHP or Html.	13 days	Tue 16/12/25	Sun 28/12/25	14
16		3.3.3 Implement the resource upload and organization features.	8 days	Mon 29/12/25	Mon 05/01/26	15
17		3.3.4 Implement flashcard generation linked to stored sources.	13 days	Tue 06/01/26	Sun 18/01/26	16
18		3.3.5 Create functionality were clicking a flashcard topic redirects users to the relevant section in the source.	8 days	Mon 19/01/26	Mon 26/01/26	17
19		3.3.6 Implement user registration and login functionality with secure authentication.	8 days	Tue 27/01/26	Tue 03/02/26	18



B.

	3.3.7 Add an admin panel to manage users, resources, and system data.	8 days	Wed 04/02/26	Wed 11/02/26	19
	Recive Feedback on Contectual report	7 days	Thu 12/02/26	Wed 18/02/26	20
	Apply Feedback given to final report	9 days	Thu 19/02/26	Fri 27/02/26	21
	Apply feedback to product development	9 days	Sat 28/02/26	Sun 08/03/26	22
	3.4 Testing				
	3.4.1 Usability testing to observe how easily students navigate the repository and flashcard features.	5 days	Mon 09/03/26	Fri 13/03/26	23
	3.4.2 Black box testing to ensure system functions work as expected for users unfamiliar with the code.	5 days	Sat 14/03/26	Wed 18/03/26	25
	3.4.3 White box testing to verify system logic and database interactions for flashcard generation and resource retrieval.	5 days	Thu 19/03/26	Mon 23/03/26	26
	3.5 Evaluation				
	3.5.1 Evaluate whether the system meets the initial functional and non-functional requirements.	5 days	Tue 24/03/26	Sat 28/03/26	27
	3.5.2 Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback.	5 days	Sun 29/03/26	Thu 02/04/26	29
	3.5.3 Discuss limitations, challenges, and potential improvements for future development.	5 days	Fri 03/04/26	Tue 07/04/26	30
	Reviewing and editing Final report	4 days	Wed 08/04/26	Sat 11/04/26	31
	Submitting Final Report	2 days	Sun 12/04/26	Mon 13/04/26	32



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